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Experiment Title : Demonstration of Signal Transmission through Wireless Media (Wi-Fi)

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Experiment No. 08: Demonstration of Signal Transmission through Wireless Media (Wi-Fi)

Theory

1. What is Wireless Media?

Wireless media refers to communication channels that transmit data without using physical cables or wires. Instead, it uses electromagnetic signals such as radio waves, infrared, or microwaves to send information between devices.

Wireless media enables devices to connect and communicate over a network without a physical connection. It is widely used in technologies such as Wi-Fi, Bluetooth, and mobile networks.

Examples of wireless media:

- Wi-Fi networks
- Bluetooth communication
- Satellite communication
- Cellular networks Advantages:
- Easy installation
- Mobility and flexibility
- Reduced wiring cost

2. Define SSID (Service Set Identifier) and its Purpose in a Wi-Fi Network

SSID (Service Set Identifier) is the name given to a wireless network that allows devices to identify and connect to a specific Wi-Fi network.

It is usually a unique name assigned to a wireless router or access point and is visible when a user searches for available Wi-Fi networks on their device.

Purpose of SSID:

- Identifies a particular wireless network
- Allows users to select the correct Wi-Fi network to connect to
- Helps differentiate between multiple wireless networks in the same area
- Provides basic network identification for authentication and security

For example, when you open Wi-Fi on your phone and see names like Home_WiFi or College_Network, those names are the SSIDs.

3. Difference Between Wired LAN and Wireless LAN (WLAN)

Feature	Wired LAN	Wireless LAN (WLAN)
Medium	Uses physical cables such as Ethernet	Uses radio waves for communication

Mobility	Limited mobility	High mobility
Installation	Requires cables and ports	Easier to install, no cables required
Speed	Usually faster and more stable	Speed may vary depending on signal strength
Security	More secure because physical access is needed	Slightly less secure if not properly protected

Example:
An Ethernet network in a computer lab is a wired LAN, while a Wi-Fi network in a home or cafe is a WLAN.

4. Significance of "Signal Strength" in Wireless Transmission

Signal strength refers to the power level of a wireless signal received by a device from a transmitter such as a router or access point.

It determines how well a device can communicate with the wireless network.

Importance of signal strength:

- Strong signal → faster and more reliable connection
- Weak signal → slower internet speed and connection drops
- Affects the coverage area of the wireless network
- Influenced by factors such as distance, walls, interference from other devices, and obstacles

For example, when you move closer to a Wi-Fi router, the signal strength increases and the internet connection becomes more stable.

Hardware Components Required

- **Wireless Router (WRT300N):** The central device that broadcasts the wireless signal and manages the network.
- **Laptop-PT (x2):** End devices equipped with wireless network interface cards.
- **Smartphone (x1):** A mobile device with built-in Wi-Fi capability.

Hardware and Connection Setup

- **Wireless Standard:** 2.4 GHz 802.11n
- **Network ID (DHCP):** 192.168.0.0 (Default for WRT300N)
- **SSID (Network Name):** MyWirelessNetwork
- **Authentication:** WPA2-Personal

Procedure

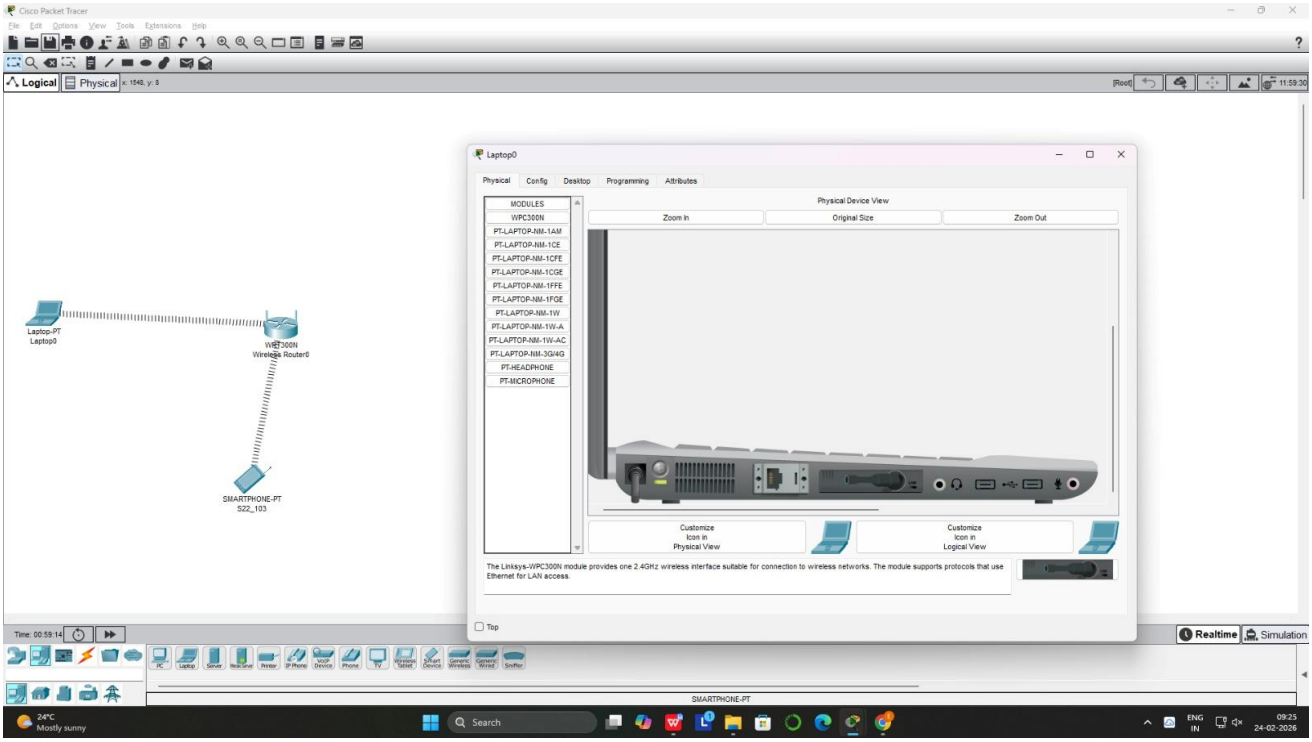
Step 1: Physical Topology Design

- Place one **Wireless Router (WRT300N)** in the center.

- Place two **Laptops** and one **Smartphone** near the router.
- Naming: Rename Smartphone following the convention (e.g., S11_01).
- **Observation:** Note that the Smartphone connects automatically (default settings), but the Laptops do not.

Step 2: Hardware Identification (Laptop Modification)

- Click on a **Laptop** > **Physical** tab.
- **Task:** Turn off the Laptop power. Remove the Ethernet module (RJ45) and drag the **WPC300N** (Wireless Module) into the slot. Turn the power back on.
- **Observation:** A dashed line appears between the Laptop and Router, indicating a wireless link.



Step 3: Wireless Router Configuration

- Click the **Wireless Router** > **Config** tab > **Wireless**.
- Set the **SSID** to MyWirelessNetwork.
- Select **WPA2-PSK** and enter a Passphrase (e.g., TSEC@2026).
- **Observation:** The dashed lines disappear because the clients now need the correct security key to connect.

Step 4: Connecting Clients to the Wireless Signal

- Click **Laptop** > **Desktop** tab > **PC Wireless**.
- Go to the **Connect** tab. Click **Refresh**. Select MyWirelessNetwork and click **Connect**.
- Enter the Passphrase TSEC@2026
- ***Repeat this for the second Laptop and the Smartphone (under Config > Wireless)***.

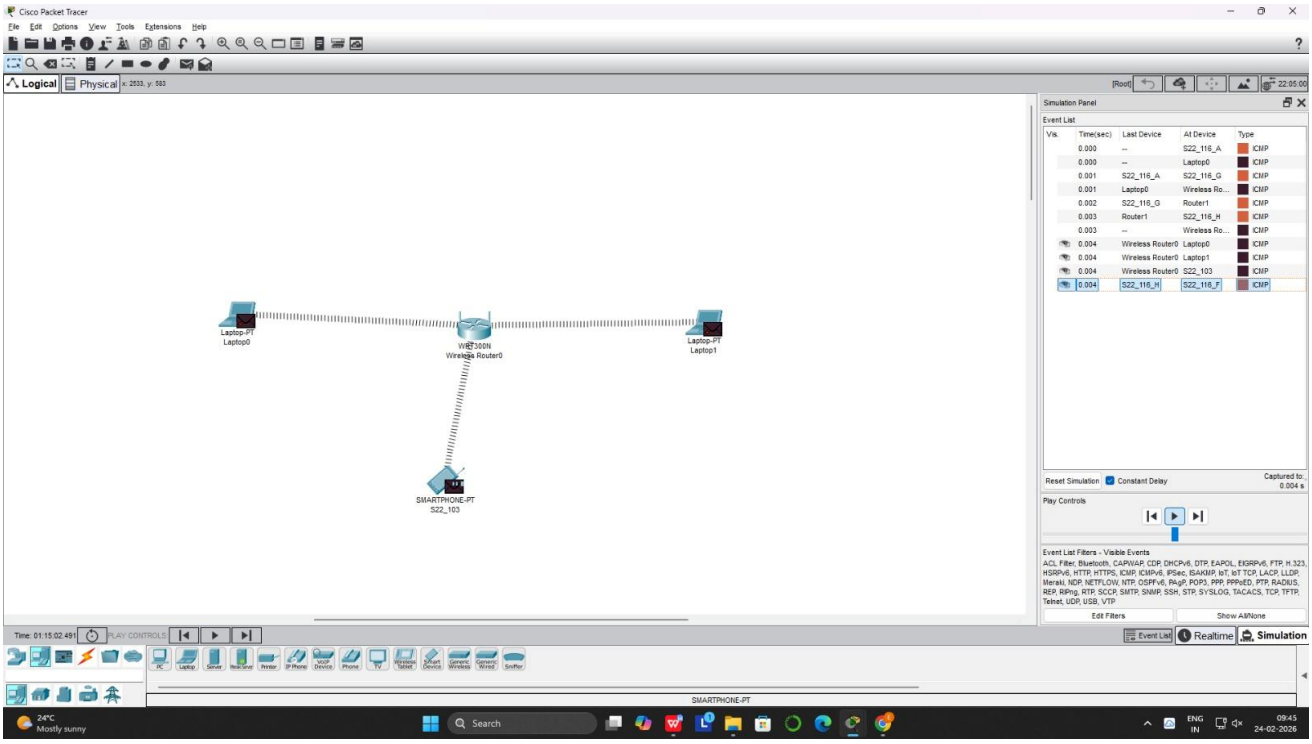
Step 5: Logical Addressing (DHCP Verification)

- Unlike the wired experiment, we will not set Static IPs.
- Click **Laptop** > **Desktop** > **IP Configuration**. Ensure **DHCP** is selected.

Device	SSID	Authenticat	Assigned IP Address
Laptop	MyWireless	WPA2-PSK	192.168.0.102
Laptop	MyWireless	WPA2-PSK	192.168.0.104
Smartp	MyWireless	WPA2-PSK	192.168.0.100

Step 6: Testing Signal Transmission

- **Real-time:**
  - Open the Command Prompt on Laptop A and ping the IP address of the Smartphone.
- **Simulation Mode:**
  - Filter for **ICMP**.
  - Use the Add Simple PDU (Envelope) to click Laptop A then Smartphone.
  - Watch the packet travel through the air (wireless media) to the Router and then to the target device.



Observations

- **No Physical Cables:** Communication was established without the use of Copper or Fiber cables, proving the use of Radio Frequency (RF) as the transmission media.
- **SSID Filtering:** Devices could only see and connect to the specific network identified by the SSID.
- **DHCP Automation:** The Wireless Router automatically managed the IP addresses, simplifying the setup of a WLAN.

Conclusion

The experiment successfully demonstrates signal transmission through **Wireless Media**. It proves that a **Wireless Router** acts as an Access Point and a DHCP server, allowing mobile and portable devices to communicate within a defined range using radio waves and security protocols.